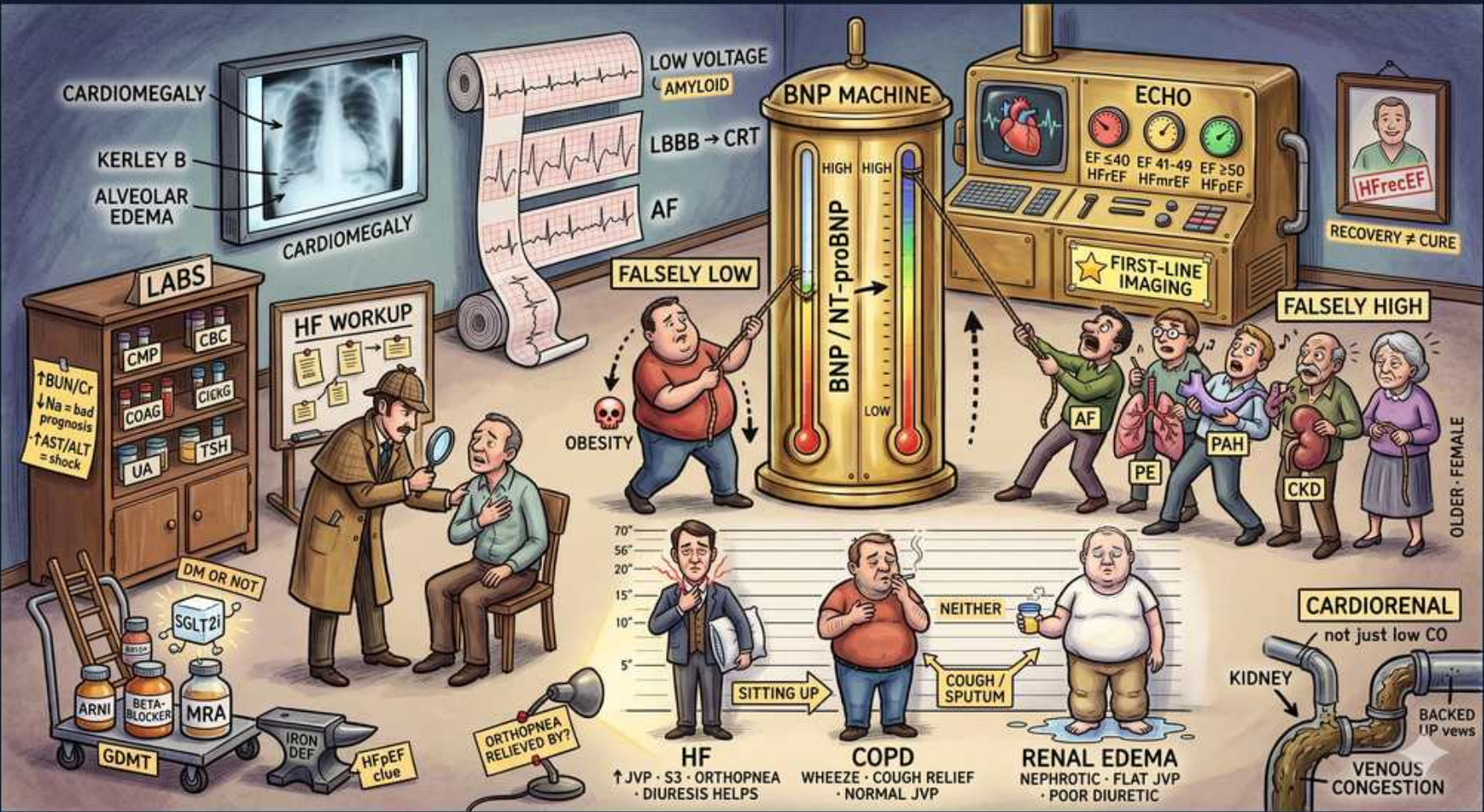


Heart Failure — SCENE 3 — HF DIAGNOSIS, BNP TRAPS & DDx



1. CXR: cardiomegaly

WHAT YOU SEE

Chest X-ray labeled CARDIOMEGALY

WHAT IT ENCODES

On chest X-ray, HF shows cardiomegaly defined as a cardiothoracic ratio >0.5 on a PA film, plus signs of pulmonary venous congestion (cephalization of vessels, perihilar 'bat-wing' edema). Cardiomegaly reflects chronic chamber dilation/remodeling and supports a cardiac cause of dyspnea.

BOARD TRAP

WRONG: a normal-sized heart on CXR rules out heart failure. Discriminator: acute HF (e.g., flash pulmonary edema from HTN or acute MI/papillary rupture) and HFpEF can present with a normal cardiac silhouette — the lungs (congestion) tell the acute story, not heart size.

2. Kerley B lines / alveolar edema

WHAT YOU SEE

X-ray annotations 'KERLEY B', 'ALVEOLAR EDEMA'

WHAT IT ENCODES

Kerley B lines are short horizontal peripheral lines from engorged interlobular septa; with alveolar/interstitial edema and pleural effusions (often right-sided) they indicate elevated pulmonary venous/wedge pressure. The radiographic congestion sequence: redistribution → interstitial (Kerley B) → alveolar edema as wedge pressure climbs above ~18–25 mmHg.

BOARD TRAP

WRONG: Kerley B lines and bilateral interstitial markings are specific for HF. Discriminator: they also occur in lymphangitic carcinomatosis, interstitial lung disease and viral pneumonitis — combine the CXR with BNP, JVP and echo before committing to HF.

3. ECG: low voltage, LBBB, AF

WHAT YOU SEE

ECG strip 'LOW VOLTAGE ± amyloid', 'LBBB → CRT', 'AF'

WHAT IT ENCODES

ECG clues in HF: low QRS voltage with a 'thick' heart on echo suggests infiltrative cardiomyopathy (cardiac amyloidosis — voltage/mass mismatch); LBBB with EF ≤35% and QRS ≥150 ms flags a CRT candidate; AF both causes (tachycardia-mediated CM) and complicates HF. Q waves point to prior infarction/ischemic etiology.

BOARD TRAP

WRONG: a normal ECG is irrelevant in suspected HF. Discriminator: a completely normal ECG has a high negative predictive value and makes systolic HF unlikely, redirecting the workup — use it as a screening clue, not a throwaway.

4. BNP / NT-proBNP machine

WHAT YOU SEE

Thermometer machine 'BNP / NT-proBNP'

WHAT IT ENCODES

Natriuretic peptides are released from stretched ventricular myocytes. Use rule-out cutoffs in acute dyspnea: BNP <100 pg/mL or NT-proBNP <300 pg/mL make HF unlikely (high NPV). Rule-IN (age-adjusted NT-proBNP): >450 (age <50), >900 (50–75), >1800 (>75). NT-proBNP is the inactive cleavage fragment with a longer half-life and renal clearance; sacubitril (ARNI) raises BNP but lowers NT-proBNP, so monitor NT-proBNP on ARNI.

BOARD TRAP

WRONG: a high BNP confirms heart failure as the cause of dyspnea. Discriminator: BNP is a sensitive RULE-OUT test (high NPV), not a specific rule-in — it rises with AF, PE, sepsis, renal failure and age, so an elevated value needs an echo to confirm cardiac dysfunction.

5. Falsely LOW: obesity

WHAT YOU SEE

Obese man under 'FALSELY LOW' arrow

WHAT IT ENCODES

Obesity FALSELY LOWERS BNP/NT-proBNP (likely via increased clearance/receptor expression in adipose tissue and altered handling). In an obese patient, a 'normal' or only mildly elevated BNP does NOT exclude HF — lower your threshold and lean on echo. This is the single most tested BNP pitfall.

BOARD TRAP

WRONG: obesity raises BNP. Discriminator: it LOWERS it — the classic board trap is the obese, dyspneic patient with a deceptively normal BNP who actually has HFpEF; don't be falsely reassured.

6. Falsely HIGH: AF, PE, PAH

WHAT YOU SEE

Crowd labeled AF, PE, PAH under 'FALSELY HIGH'

WHAT IT ENCODES

BNP/NT-proBNP rise from any cause of myocardial stretch or RV strain, not just LV failure: atrial fibrillation, pulmonary embolism, pulmonary arterial hypertension, sepsis, high-output states. In PE, an elevated BNP/troponin marks RV strain and worse prognosis — but it is not diagnostic of LV HF.

BOARD TRAP

WRONG: an elevated BNP in a dyspneic patient means the cause is left heart failure, so anticoagulation for PE can wait. Discriminator: AF, PE and PAH all raise BNP via RV/atrial stretch — a mildly raised BNP must not anchor you onto HF and away from a missed PE.

7. Falsely HIGH: CKD, older/female

WHAT YOU SEE

Elderly woman, 'CKD', 'OLDER / FEMALE'

WHAT IT ENCODES

BNP and especially NT-proBNP rise with renal impairment (reduced clearance — NT-proBNP is renally cleared and most affected), with advancing age, and with female sex. This is exactly why NT-proBNP cutoffs are AGE-ADJUSTED (>450 / >900 / >1800). In CKD, use higher rule-in thresholds (e.g., NT-proBNP >1200) before attributing dyspnea to HF.

BOARD TRAP

WRONG: renal failure lowers NT-proBNP. Discriminator: it RAISES it because NT-proBNP is cleared by the kidney — a high value in a CKD patient may reflect reduced clearance, not new HF, so interpret against an age-/renal-adjusted threshold.

8. Echo = first-line imaging

WHAT YOU SEE

ECHO screen showing EF cutoffs, 'FIRST-LINE IMAGING'

WHAT IT ENCODES

Transthoracic echocardiography is the FIRST-LINE imaging test in HF: it gives the LVEF (defining HFrEF ≤ 40 , HFmrEF 41–49, HFpEF ≥ 50), wall motion, chamber size, diastolic function (E/e'), valve disease, pericardium and estimated PA pressures. Cardiac MRI is reserved for tissue characterization (amyloid, sarcoid, myocarditis, scar/viability).

BOARD TRAP

WRONG: a normal LVEF on echo excludes heart failure. Discriminator: HFpEF has EF $\geq 50\%$ with abnormal diastolic indices (raised E/e', LA enlargement) — you must look at diastology and filling pressures, not just the ejection fraction number.

9. Lab workup (TSH, iron, etc.)

WHAT YOU SEE

Cabinet 'HF WORKUP' with TBili, CBC, LFTs, TSH, Fe folders

WHAT IT ENCODES

Baseline HF labs hunt for reversible drivers and comorbidities: CBC (anemia), electrolytes/renal function (baseline before RAAS/MRA and to track), LFTs/bilirubin (congestive hepatopathy), TSH (thyroid CM is reversible), iron studies (ferritin + transferrin sat — deficiency is common and treatable with IV iron), HbA1c/glucose, and natriuretic peptides.

BOARD TRAP

WRONG: checking TSH and iron studies in HF is low-yield box-ticking. Discriminator: both thyroid disease and iron deficiency are reversible/treatable contributors — missing thyrotoxic or hypothyroid cardiomyopathy, or iron deficiency amenable to IV iron, is a real management miss.

10. DDx — HF (JVP, orthopnea, diuretics help)

WHAT YOU SEE

Man relieved sitting up, '↑JVP, S3, orthopnea, diuresis helps'

WHAT IT ENCODES

HF dyspnea pattern: elevated JVP, an S3 gallop (volume overload, the most specific bedside sign of HFrEF), orthopnea and paroxysmal nocturnal dyspnea, bilateral basal crackles and edema, and clear symptomatic relief with diuresis. The most specific findings are S3 and raised JVP.

BOARD TRAP

WRONG: peripheral edema by itself diagnoses heart failure. Discriminator: edema is nonspecific (venous insufficiency, nephrotic syndrome, cirrhosis, calcium-channel blockers) — it is the COMBINATION with elevated JVP, S3 and orthopnea that points to HF.

11. DDx — COPD (normal JVP)

WHAT YOU SEE

Man 'COPD: wheeze, cough relief, NORMAL JVP'

WHAT IT ENCODES

COPD/asthma dyspnea: expiratory wheeze, prolonged expiration, cough with sputum, hyperinflation on CXR, NORMAL JVP, and a BNP that is typically normal or only mildly raised. Relief comes from bronchodilators, not diuresis. (Cor pulmonale from advanced COPD can raise JVP and BNP via RV strain — a later overlap.)

BOARD TRAP

WRONG: wheezing means asthma/COPD, so it cannot be heart failure. Discriminator: 'cardiac asthma' (HF can wheeze from peribronchial edema) — check JVP and BNP; normal JVP + normal BNP favors primary lung disease, raised both favor HF.

12. DDx — renal edema (poor diuretic response)

WHAT YOU SEE

Man 'RENAL EDEMA: nephrotic, flat JVP, poor diuretic'

WHAT IT ENCODES

Renal/nephrotic edema: heavy proteinuria, hypoalbuminemia, FLAT JVP, normal cardiac output, and a relatively poor/blunted response to loop diuretics (low oncotic pressure and avid sodium retention). Distinguished from HF by the flat JVP and the absence of S3/orthopnea.

BOARD TRAP

WRONG: bilateral leg edema with poor diuretic response must be decompensated heart failure. Discriminator: a FLAT JVP points away from HF toward renal/hepatic/hypoalbuminemic edema — check urine protein and albumin rather than escalating cardiac therapy.

13. Cardiorenal syndrome

WHAT YOU SEE

Kidney + tap 'CARDIORENAL — not just low CO / venous congestion'

WHAT IT ENCODES

Cardiorenal syndrome is renal dysfunction in HF driven largely by elevated central VENOUS pressure/congestion (renal venous congestion lowering net glomerular perfusion), not merely by low forward cardiac output. The counterintuitive treatment is often decongestion (continue/intensify diuresis) rather than holding diuretics for a rising creatinine.

BOARD TRAP

WRONG: a creatinine bump during HF diuresis means you are over-diuresing and should stop the loop diuretic. Discriminator: in congestive cardiorenal syndrome continued decongestion frequently IMPROVES renal function — venous congestion, not volume depletion, is the driver; assess volume status before pulling back.

14. GDMT drugs at base

WHAT YOU SEE

Bottles 'ARNI, BETA, MRA, SGLT2' under GDMT

WHAT IT ENCODES

Once HFrEF is diagnosed (EF $\leq 40\%$), start the FOUR pillars early and titrate: ARNI sacubitril-valsartan (preferred over ACEi/ARB), an evidence-based beta-blocker (carvedilol, metoprolol succinate, or bisoprolol), an MRA (spironolactone/epiarenone), and an SGLT2i (dapagliflozin/empagliflozin). Current practice initiates all four rapidly (in parallel) rather than sequentially over months.

BOARD TRAP

WRONG: titrate one HFrEF pillar to its target dose before adding the next class. Discriminator: modern strategy (STRONG-HF) favors rapid simultaneous initiation of all four classes at low doses then up-titration — early quadruple therapy reduces mortality and readmissions versus slow sequential dosing.